Honest ADMET: When the Generalization Gap Appears, Why, and Whether Selective Prediction Recovers It

Anonymous Author(s) anonymous@ml4h.cc

Editor: Editors of Machine Learning for Health

Abstract

Public ADMET leaderboards such as Therapeutics Data Commons (TDC) rank models by a single score under one fixed split. We ask three questions a deployer actually faces, across all 22 TDC ADMET endpoints: (i) *how much* does measured performance fall under realistic, chemistry-aware splits versus an optimistic random split; (ii) *why* the effect is heterogeneous and sometimes reverses; and (iii) whether *selective prediction* recovers the reliability that random-split numbers overstate. We evaluate two molecular representations (ECFP4 fingerprints and a frozen pretrained-transformer embedding) across all 22 TDC ADMET endpoints. We make three contributions. First, a **statistically honest gap atlas**: signed, per-endpoint relative gaps with matched-replicate, split-seed block-bootstrap CIs, paired Wilcoxon tests, and Benjamini–Hochberg FDR. Bemis–Murcko *scaffold* splits are much weaker stressors than *cluster* (sphere-exclusion) splits for ADMET—their median relative gap is $\sim$2.3$\times$ smaller (+0.10 vs +0.23) and not established on 5/22 endpoints including hERG, consistent with Fooladi et al. (2025)—while cluster splits reveal a gap established on **21/22 endpoints** (block-bootstrap CIs; aggregate $p < 10^{-4}$). Second, error spikes $\sim$8$\times$ in the most dissimilar similarity bin, so the gap is an applicability-domain effect. Third, we report an **honest evaluation of uncertainty under shift, negatives included**: split-conformal under-covers under cluster shift, covariate-shift weighting is a verified no-op that does not restore coverage, the best abstention signal is model-dependent, and per-endpoint abstention benefit *does not* track the optimism gap ($\rho = 0.07$, n.s.). Frozen ChemBERTa embeddings shrink the gap modestly but significantly ($p = 0.011$) without closing it. All re-split numbers are anchored to TDC's official-split fingerprint baselines first; code and splits are released as a `pip`-installable, leakage-audited harness.

Keywords: ADMET; distribution shift; conformal prediction; selective prediction; benchmark evaluation; Therapeutics Data Commons

1. Introduction

ADMET (absorption, distribution, metabolism, excretion, toxicity) prediction is where much of early drug discovery succeeds or fails, and Therapeutics Data Commons (TDC) (Huang et al., 2021, 2022) is the community-standard benchmark for it. Its leaderboard culture optimizes a single metric under one fixed scaffold split—useful for ranking, but answering a different question than the one a practitioner faces: *if I deploy this model on genuinely novel chemistry, how well will it work, and will it tell me when it does not know?*

A long line of work shows random splits leak structurally similar molecules across folds and overstate prospective performance (Sheridan, 2013; Wu et al., 2018; Deng et al., 2023). Scaffold splits were introduced as a "harder, more realistic" default (Wu et al., 2018). Yet recent results complicate this: Fooladi et al. (2025) find that for ADMET, Bemis–Murcko scaffold splits are *not substantially harder than random*, and only cluster/distance splits

Table 1: Differentiation from the closest prior work. Y = yes, P = partial, – = no.

Work	#Endpts	Model fams	Conformal	Ensemble	Realistic split	Selective-pred	Found. model
CFR (Li et al., 2024)	P	1	Y	Y	P	P	–
CoDrug (Laghuvarapu et al., 2023)	P	1	Y	–	Y	–	–
MOOD (Tossou et al., 2024)	P	P	–	Y	Y	–	–
Fooladi (Fooladi et al., 2025)	P	Y	–	–	Y	–	–
Koleiev (Koleiev et al., 2026)	Y (22)	Y	–	–	P	–	P
This work	**Y (22)**	**P**	**Y**	**Y**	**Y**	**Y**	**Y**

expose a real gap; Guo et al. (2024) show scaffold splits still leak because different scaffolds can be near-identical molecules. Separately, the uncertainty-quantification literature offers conformal prediction (Tibshirani et al., 2019; Angelopoulos and Bates, 2023), deep ensembles (Scalia et al., 2020; Hirschfeld et al., 2020), and conformal-under-shift methods (Laghuvarapu et al., 2023; Li et al., 2024)—but these are rarely tied back to the split-induced optimism they could mitigate.

We are explicit that *neither phenomenon is a new discovery*. Our contribution is a rigorous *consolidation and extension* that no single prior paper provides, and—in the spirit of Kapoor and Narayanan (2023)—one that reports negatives plainly. This extends the trustworthy-ML-under-shift agenda of the Zitnik lab's own tools (TDC → TxGNN (Huang et al., 2024), the last of which defers uncertainty quantification to future work).

Contributions. (1) A statistically honest, per-endpoint **generalization-gap atlas** across 22 endpoints and model families, engaging the scaffold≈random result head-on and promoting cluster splits to the primary stressor. (2) A mechanistic **applicability-domain** account of the gap. (3) An **honest evaluation of uncertainty under shift**, including clean negative results, plus a foundation-model arm asking whether pretrained representations close the gap.

2. Related Work

Split realism. Sheridan (2013) established that train–test dissimilarity governs apparent accuracy; Wu et al. (2018) popularized scaffold splits; Deng et al. (2023) confirmed scaffold < random performance. Guo et al. (2024) and Fooladi et al. (2025) argue scaffold splits under-stress and cluster/distance splits are needed; van Tilborg et al. (2022) expose a complementary within-scaffold failure (activity cliffs). **TDC and critique.** TDC (Huang et al., 2021, 2022) is our object of study; concurrent work (Koleiev et al., 2026) audits the 22 ADMET leaderboards for leakage and reproducibility—complementary to our controlled signed-gap measurement. **Uncertainty / selective prediction.** We *apply* (not invent) split conformal (Tibshirani et al., 2019; Angelopoulos and Bates, 2023), deep ensembles (Scalia et al., 2020; Hirschfeld et al., 2020), and conformal-under-covariate-shift (Laghuvarapu et al., 2023; Li et al., 2024), evaluated via risk–coverage curves (Geifman and El-Yaniv, 2017). **Foundation models.** We evaluate frozen ChemBERTa (Ahmad et al., 2022) (cf. MolFormer (Ross et al., 2022)). Table 1 differentiates this work from the closest prior studies.

3. Data and Methods

Datasets. All 22 endpoints of the TDC ADMET benchmark group (9 regression, 13 classification), via PyTDC 1.1.15 with pinned RDKit 2023.9.6. **Two loading modes.** For leaderboard reproduction we use TDC's native, unpooled split and the official 5-seed `admet_group` evaluate protocol. For the controlled split-type comparison we pool train+test, canonicalize, de-duplicate on canonical SMILES, and re-split under explicit seeds; pooled numbers are internal-comparison-only. **Splits.** *random* (optimistic baseline); *scaffold* (Bemis–Murcko, acyclic molecules treated as singleton groups, not lumped); *cluster* (sphere-exclusion via RDKit LeaderPicker on ECFP4—our primary OOD stressor, $O(N \cdot L)$ and memory-safe on the ∼13k-molecule CYP sets). **Leakage** is reported with a splitter-independent nearest-neighbour ECFP4 Tanimoto metric. **Models.** ECFP4 + XGBoost baselines for the atlas, with a random-forest robustness spot-check on 5 endpoints; and a frozen ChemBERTa-embedding arm with an XGBoost head (a representation swap, not a new model class). A GNN baseline is left to future work.

Generalization gap and statistics. For each (dataset, model) and realistic split we compute the gap per *matched replicate* (same split seed and model seed for random and realistic, so the comparison is paired). We report the mean gap, a 95% bootstrap CI over replicates, a paired Wilcoxon p-value, and Benjamini–Hochberg q-values across the 22-endpoint family; a gap whose CI crosses zero is "not established." An aggregate Wilcoxon over per-endpoint gaps gives the headline claim.

Conformal and selective prediction. Split-conformal with the validation fold as calibration: absolute-residual intervals (regression) and LAC sets (classification). We implement weighted conformal under covariate shift (Tibshirani et al., 2019) with importance weights from an isotonic-calibrated, class-balanced domain classifier, clipped at the 99th percentile, with an effective-sample-size fallback. Our quantile reduces *exactly* to textbook split-conformal under uniform weights (unit-tested) and returns $+\infty$ when coverage cannot be certified. Selective prediction uses risk–coverage curves with two label-free signals—deep-ensemble disagreement and applicability-domain (AD) similarity—compared by Excess-AURC (AURC minus the oracle ordering), which is comparable across folds with different base error rates.

4. Results

4.1. The harness matches untuned fingerprint baselines

Under the official 5-seed protocol, our deliberately simple ECFP4+XGBoost baselines land in the range of TDC's untuned fingerprint baselines (Table 2, full list in Appendix). They are *not* competitive with tuned leaderboard entries (e.g. our Caco-2 MAE 0.386 vs. a tuned XGBoost leaderboard entry near 0.27); the point is only to confirm the harness is sane and in-range before any re-split is introduced, so the subsequent re-split numbers are interpretable rather than an artifact.

4.2. Scaffold under-stresses; cluster reveals a universal gap

Across all 22 endpoints (8 split seeds × 3 model seeds, matched-replicate *relative* gaps = gap/random-split score; CIs from a split-seed block bootstrap that resamples the unit of

Table 2: Leaderboard reproduction (official TDC split, 5-seed protocol), excerpt.

Endpoint	Metric	ECFP4+XGBoost (mean±std)
caco2_wang	MAE	0.386 ± 0.030
bbb_martins	AUROC	0.872 ± 0.006
herg	AUROC	0.727 ± 0.008
solubility_aqsoldb	MAE	1.254 ± 0.009

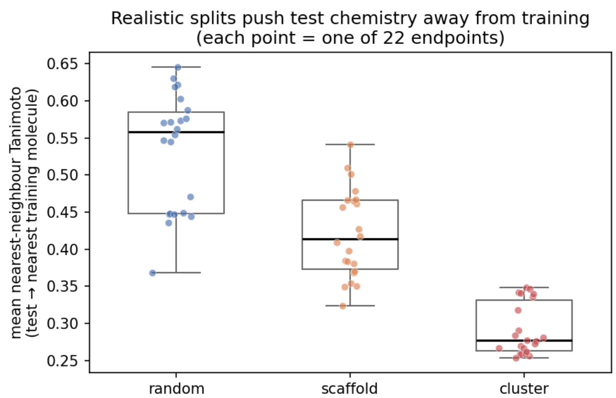

Figure 1: Mean nearest-neighbour Tanimoto similarity (test → nearest training molecule) per endpoint. Realistic splits push the test set out of the training domain; cluster is the strongest stressor.

independence), under *cluster* splits 21/22 endpoints show a statistically established gap (95% CI excludes zero), with median relative gap +0.234 and aggregate Wilcoxon $p < 10^{-4}$. Under *scaffold* splits gaps are markedly smaller (median +0.102) and 5/22 are not established; most strikingly hERG is flat under scaffold (−0.002, n.s.) but a clear +0.066 under cluster. The cluster effect is ≈2.3× the scaffold median (Figure 2; per-endpoint Appendix table with block-bootstrap CIs and per-split established flags). Because raw-unit gaps are not cross-endpoint comparable (PPBR's 0–100% target gives a large MAE gap), we report relative gaps throughout. The realistic splits genuinely push test chemistry out of domain: mean nearest-neighbour Tanimoto similarity to the training set falls from 0.56 (random) to 0.41 (scaffold) to 0.28 (cluster) across the 22 endpoints (Figure 1).

4.3. The gap is an applicability-domain effect

Binning every test prediction by nearest-neighbour Tanimoto similarity to training, normalized error *spikes* to ≈7.9× the endpoint mean in the most dissimilar bin (similarity < 0.1, a sparsely populated tail) and is already near-baseline (≈1.3×) by the next decile (Figure 3). Error is concentrated in molecules far from training—explaining the gap and motivating distance as an abstention signal. Across endpoints, the drop in mean similarity from random to cluster splits is suggestively associated with the relative gap (Spearman $\rho = +0.41$, $p = 0.06$). We flag a shared-representation caveat: ECFP4 Tanimoto defines

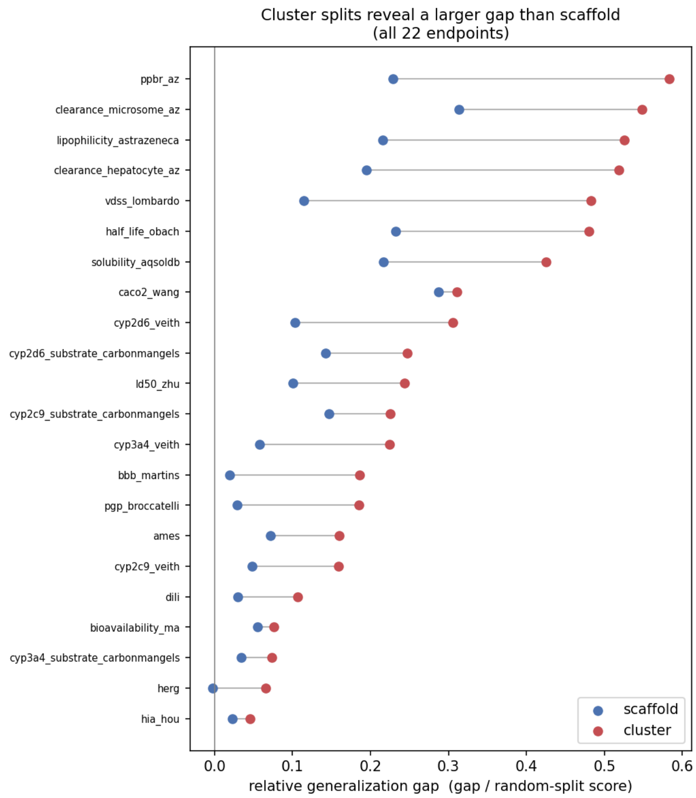

Figure 2: Relative generalization gap per endpoint: cluster (red) exceeds scaffold (blue) on all 22 endpoints. hERG, flat under scaffold, shows a positive cluster gap.

the cluster split, the leakage metric, *and* this AD signal, so the relationship is partly built in; a splitter-independent distance control is left to future work.

4.4. Uncertainty under shift: an honest accounting

Corrected split-conformal achieves nominal coverage on exchangeable data (self-test 0.901 at the 0.90 target). On real splits, coverage degradation under cluster shift is *endpoint-dependent and modest in aggregate*: mean cluster coverage is 0.896 (vs. 0.907 under random; target 0.90), but it drops sharply on specific high-shift endpoints (BBB 0.92 → 0.84; 4/22 endpoints below 0.88; Figure 4). Covariate-shift weighting is correctly a no-op on random splits but did *not* restore cluster coverage: the importance weights stayed near-uniform (Kish ESS ≈0.90), consistent with the known weakness of classifier density-ratios under high-dimensional shift; we report ESS and fall back rather than over-claim.

For abstention, a random-forest pilot suggested AD distance beat ensemble disagreement under cluster shift; *this did not replicate* with XGBoost ensembles, where ensemble disagreement is significantly better (wins on 18/22 endpoints, paired Wilcoxon $p = 0.007$) and AD abstention actively *hurts* on 9/22. The headline is thus the cross-model *instability*

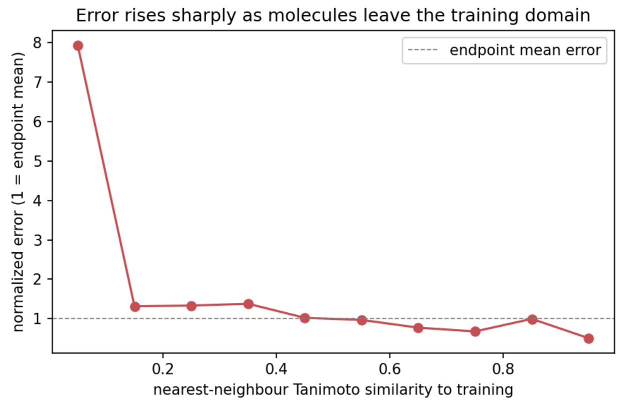

Figure 3: Normalized error vs. nearest-neighbour similarity to training (mean over 22 endpoints).

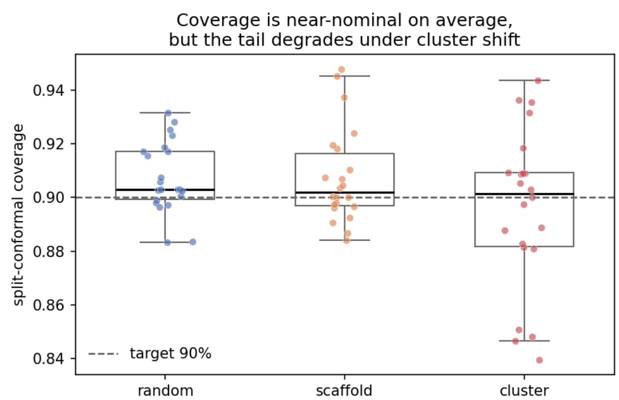

Figure 4: Split-conformal coverage per endpoint (target 0.90). Near-nominal on average for all splits, but the lower tail degrades under cluster shift (worst: BBB ≈0.84); degradation is endpoint-specific, not uniform.

of the best abstention signal, not a winner. Finally, we find no evidence that high-gap endpoints benefit more from abstention (Spearman $\rho = +0.07$, 95% CI $[-0.36, +0.48]$, under-powered at n=22; Figure 5): selective prediction does not cleanly "recover the gap."

4.5. Do foundation models close the gap? Modestly

Running the same machinery on frozen ChemBERTa embeddings + an XGBoost head, the foundation model shrinks the relative cluster gap versus ECFP4 on 17/22 endpoints (mean $\Delta = -0.014$, paired Wilcoxon $p = 0.011$), but the effect is small and non-uniform, and the gap remains large (Figure 6; Table in Appendix). Pretrained representations help but do *not* solve out-of-distribution generalization.

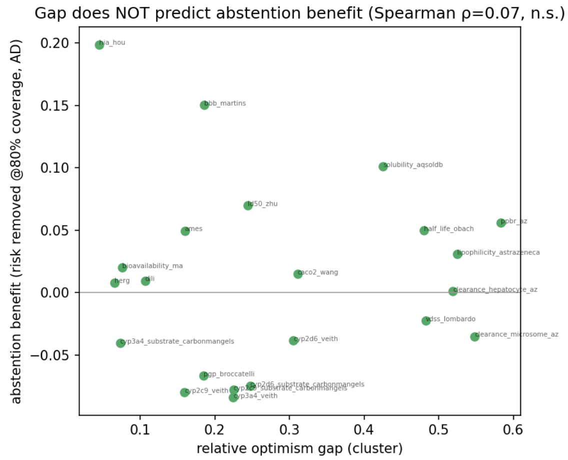

Figure 5: Per-endpoint optimism gap vs. abstention benefit under cluster shift: no correlation ($\rho = 0.07$, n.s.). An honest null.

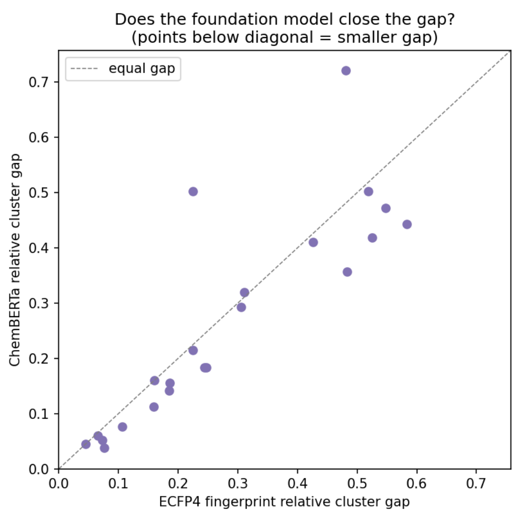

Figure 6: ChemBERTa vs. ECFP4 relative cluster gap. Points below the diagonal = smaller gap.

5. Discussion

The message is not "models are bad" but "the standard evaluation is optimistic in a *predictable* way": measured performance tracks how far test chemistry sits from training, and scaffold splits under-state it. The natural remedy—abstain on the OOD tail—is more fragile than one would hope: conformal itself under-covers under the same shift, the best abstention signal is model-dependent, and abstention benefit does not track the gap. The practical takeaway is to **report cluster-split performance and nearest-neighbour applicability domain as standard**, and to treat uncertainty methods as helpful but shift-sensitive rather than a turnkey fix. Our negative results are prescriptive: classifier density-ratio covariate-shift correction is unreliable for ADMET fingerprints (ESS collapse) and should not be relied upon, and the best abstention signal is model-dependent—there is no turnkey uncertainty signal, so it must be validated per model class.

6. Limitations

Compute was bounded to a single consumer device (Apple M4 Pro), which keeps the study fully reproducible on commodity hardware but limits the foundation-model arm to frozen embeddings; GNNs and fine-tuning are future work. Classifier density-ratio weights collapse in ESS under severe shift (we report ESS and fall back). Temporal splits are out of scope for most endpoints (no provenance dates) and not claimed. Pooled re-splits are not leaderboard-comparable by construction; we anchor with the official-split reproduction.

7. Conclusion

A reproducible, leakage-audited re-evaluation of TDC ADMET shows *when* and *why* the generalization gap appears—universally under cluster splits, driven by distance-to-training—and gives an honest accounting of what uncertainty methods do and do not buy under shift. Code, pinned environment, splits, and figures are released.

References

Walid Ahmad, Elana Simon, Seyone Chithrananda, Gabriel Grand, and Bharath Ramsundar. Chemberta-2: towards chemical foundation models. *arXiv preprint*, 2022. arXiv:2209.01712.

Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *Foundations and Trends in Machine Learning*, 2023. arXiv:2107.07511.

Jianyuan Deng, Zhibo Yang, Hehe Wang, Iwao Ojima, Dimitris Samaras, and Fusheng Wang. A systematic study of key elements underlying molecular property prediction. *Nature Communications*, 14:6395, 2023. doi: 10.1038/s41467-023-41948-6.

Hosein Fooladi, Thi Ngoc Lan Vu, Miriam Mathea, and Johannes Kirchmair. Evaluating machine learning models for molecular property prediction: performance and robustness on out-of-distribution data. *Journal of Chemical Information and Modeling*, 65(19):9871–9891, 2025. doi: 10.1021/acs.jcim.5c00475.

Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Qianrong Guo, Saiveth Hernandez-Hernandez, and Pedro J. Ballester. Scaffold splits overestimate virtual screening performance. In *International Conference on Artificial Neural Networks (ICANN)*, 2024. arXiv:2406.00873.

Lior Hirschfeld, Kyle Swanson, Kevin Yang, Regina Barzilay, and Connor W. Coley. Uncertainty quantification using neural networks for molecular property prediction. *Journal of Chemical Information and Modeling*, 60(8):3770–3780, 2020. doi: 10.1021/acs.jcim.0c00502.

Kexin Huang, Tianfan Fu, Wenhao Gao, et al. Therapeutics data commons: machine learning datasets and tasks for drug discovery and development. In *NeurIPS Datasets and Benchmarks*, 2021. arXiv:2102.09548.

Kexin Huang, Tianfan Fu, Wenhao Gao, et al. Artificial intelligence foundation for therapeutic science. *Nature Chemical Biology*, 18:1033–1036, 2022. doi: 10.1038/s41589-022-01131-2.

Kexin Huang, Payal Chandak, Qianwen Wang, others, and Marinka Zitnik. A foundation model for clinician-centered drug repurposing. *Nature Medicine*, 30:3601–3613, 2024. doi: 10.1038/s41591-024-03233-x.

Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804.

[verify] Koleiev et al. Critical assessment of machine learning models for admet prediction in tdc leaderboards. *bioRxiv*, 2026. doi: 10.64898/2026.02.26.708193. Author list and DOI to be verified.

Siddhartha Laghuvarapu, Zhen Lin, and Jimeng Sun. Codrug: conformal drug property prediction with density estimation under covariate shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2310.12033.

[verify] Li et al. Conformalized graph learning for molecular admet property prediction and reliable uncertainty quantification. *Journal of Chemical Information and Modeling*, 64(23):8705–8717, 2024. doi: 10.1021/acs.jcim.4c01139.

Jerret Ross, Brian Belgodere, Vijil Chenthamarakshan, et al. Large-scale chemical language representations capture molecular structure and properties. *Nature Machine Intelligence*, 4:1256–1264, 2022. doi: 10.1038/s42256-022-00580-7.

Gabriele Scalia, Colin A. Grambow, Barbara Pernici, Yi-Pei Li, and William H. Green. Evaluating scalable uncertainty estimation methods for deep learning-based molecular property prediction. *Journal of Chemical Information and Modeling*, 60(6):2697–2717, 2020. doi: 10.1021/acs.jcim.9b00975.

Robert P. Sheridan. Time-split cross-validation as a method for estimating the goodness of prospective prediction. *Journal of Chemical Information and Modeling*, 53(4):783–790, 2013. doi: 10.1021/ci400084k.

Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. arXiv:1904.06019.

Prudencio Tossou et al. Real-world molecular out-of-distribution: specification and investigation. *Journal of Chemical Information and Modeling*, 2024. doi: 10.1021/acs.jcim.3c01774.

Derek van Tilborg, Alisa Alenicheva, and Francesca Grisoni. Exposing the limitations of molecular machine learning with activity cliffs. *Journal of Chemical Information and Modeling*, 62(23): 5938–5951, 2022. doi: 10.1021/acs.jcim.2c01073.

Zhenqin Wu, Bharath Ramsundar, Evan N. Feinberg, et al. Moleculenet: a benchmark for molecular machine learning. *Chemical Science*, 9(2):513–530, 2018. doi: 10.1039/c7sc02664a.

Appendix A. Full leaderboard reproduction (all 22 endpoints)

Endpoint	Metric	ECFP4+XGBoost (mean±std)
ames	auroc	0.832 ± 0.001
bbb_martins	auroc	0.872 ± 0.006
bioavailability_ma	auroc	0.693 ± 0.006
caco2_wang	mae	0.386 ± 0.030
clearance_hepatocyte_az	spearman	0.316 ± 0.017
clearance_microsome_az	spearman	0.425 ± 0.021
cyp2c9_substrate_carbonmangels	auprc	0.371 ± 0.012
cyp2c9_veith	auprc	0.753 ± 0.003
cyp2d6_substrate_carbonmangels	auprc	0.633 ± 0.016
cyp2d6_veith	auprc	0.633 ± 0.004
cyp3a4_substrate_carbonmangels	auroc	0.641 ± 0.010
cyp3a4_veith	auprc	0.852 ± 0.002
dili	auroc	0.857 ± 0.004
half_life_obach	spearman	0.301 ± 0.028
herg	auroc	0.727 ± 0.008
hia_hou	auroc	0.955 ± 0.002
ld50_zhu	mae	0.634 ± 0.011
lipophilicity_astrazeneca	mae	0.666 ± 0.004
pgp_broccatelli	auroc	0.908 ± 0.004
ppbr_az	mae	10.270 ± 0.224
solubility_aqsoldb	mae	1.254 ± 0.009
vdss_lombardo	spearman	0.401 ± 0.016

Appendix B. Per-endpoint generalization-gap atlas

Relative gaps (gap / random-split score); cluster column with 95% bootstrap CI and BH-FDR q.

Endpoint	Metric	Scaffold	Cluster	Cluster 95% CI (q)
ppbr_az	mae	0.229	0.583	[+5.614, +5.909] (q0.000)
clearance_microsome_az	spearman	0.314	0.548	[+0.261, +0.304] (q0.000)
lipophilicity_astrazeneca	mae	0.216	0.525	[+0.307, +0.318] (q0.000)
clearance_hepatocyte_az	spearman	0.195	0.519	[+0.222, +0.273] (q0.000)
vdss_lombardo	spearman	0.114	0.483	[+0.219, +0.258] (q0.000)
half_life_obach	spearman	0.232	0.480	[+0.106, +0.186] (q0.000)
solubility_aqsoldb	mae	0.216	0.425	[+0.404, +0.425] (q0.000)
caco2_wang	mae	0.287	0.311	[+0.092, +0.111] (q0.000)
cyp2d6_veith	auprc	0.103	0.305	[+0.204, +0.219] (q0.000)
cyp2d6_substrate_carbonmangels	auprc	0.143	0.247	[+0.107, +0.196] (q0.000)
ld50_zhu	mae	0.101	0.244	[+0.108, +0.118] (q0.000)
cyp2c9_substrate_carbonmangels	auprc	0.147	0.225	[+0.058, +0.102] (q0.000)
cyp3a4_veith	auprc	0.058	0.225	[+0.188, +0.199] (q0.000)
bbb_martins	auroc	0.020	0.186	[+0.164, +0.177] (q0.000)
pgp_broccatelli	auroc	0.029	0.185	[+0.166, +0.177] (q0.000)
ames	auroc	0.072	0.160	[+0.137, +0.147] (q0.000)
cyp2c9_veith	auprc	0.049	0.159	[+0.119, +0.133] (q0.000)
dili	auroc	0.030	0.106	[+0.069, +0.111] (q0.000)
bioavailability_ma	auroc	0.055	0.076	[+0.035, +0.073] (q0.000)
cyp3a4_substrate_carbonmangels	auroc	0.034	0.073	[+0.030, +0.066] (q0.000)
herg	auroc	-0.002	0.066	[+0.043, +0.068] (q0.000)
hia_hou	auroc	0.022	0.045	[+0.016, +0.070] (q0.010)

Appendix C. Foundation model vs. fingerprint (relative cluster gap)

Endpoint	ECFP4	ChemBERTa	Δ
ppbr_az	0.583	0.443	-0.140
vdss_lombardo	0.483	0.357	-0.125
lipophilicity_astrazeneca	0.525	0.419	-0.106
clearance_microsome_az	0.548	0.473	-0.075
cyp2d6_substrate_carbonmangels	0.247	0.184	-0.064
ld50_zhu	0.244	0.183	-0.061
cyp2c9_veith	0.159	0.112	-0.047
pgp_broccatelli	0.185	0.142	-0.044
bioavailability_ma	0.076	0.038	-0.038
bbb_martins	0.186	0.155	-0.031
dili	0.106	0.076	-0.030
cyp3a4_substrate_carbonmangels	0.073	0.052	-0.021
clearance_hepatocyte_az	0.519	0.503	-0.016
solubility_aqsoldb	0.425	0.410	-0.015
cyp2d6_veith	0.305	0.293	-0.013
cyp3a4_veith	0.225	0.215	-0.010
herg	0.066	0.060	-0.006
hia_hou	0.045	0.045	+0.000
ames	0.160	0.160	+0.001
caco2_wang	0.311	0.320	+0.009
half_life_obach	0.480	0.721	+0.241
cyp2c9_substrate_carbonmangels	0.225	0.502	+0.277
Mean	0.281	0.267	-0.014